CPE 420: Intro to Data Mining

HomeWork 3

By:

Hala Almutairi 2211112873

Supervised by:

Dr. Ahmed Nasrallah
Eng. Anwar Al-Ebraheem

College of Engineering and Petroleum
Kuwait University
Spring 2025/2026

AI Usage :

Q4] AI was used to help understand the differences between eager learning and lazy learning more than what the slide material explains.

1. (4 points) Consider a binary classification problem with the following set of attributes and attribute values:

- Air Conditioner = {Working, Broken}
- Engine = {Good, Bad}
- Mileage = {High, Medium, Low}
- Rust = {Yes, No}

Suppose a rule-based classifier produces the following rule set:

Rule Condition	Value
Mileage = High	Low
Mileage = Low	High
Air Conditioner = Working, Engine = Good	High
Air Conditioner = Working, Engine = Bad	Low
Air Conditioner = Broken	Low

(a) Are the rules mutually exclusive? No , One car can match multiple rules

(b) Is the rule set exhaustive? Yes ,Every car matches at least one rule

(c) Is ordering needed for this set of rules? Yes , because of conflicts and not mutually exclusive

(d) Do you need a default class for the rule set? No, Every car matches at least one rule

2. (5 points) Consider the following training set for predicting whether there is traffic congestion in the morning on a highway for a particular day. There are 100 examples in the training set, with 40% positive (congestion) and 60% negative (no congestion) examples.

Accident	Weather	Construction	Positive	Negative
No	Good	No	5	30
No	Good	Yes	10	20
Yes	Good	No	10	5
Yes	Bad	No	10	5
Yes	Bad	Yes	5	0

(a) Compute the following class-conditional probabilities using the training data:

P(Accident = Yes | +) = 25 / 40 = 0.625
P(Accident = Yes | −) = 10/ 60 = 0.1667
P(Weather = Good | +) = 25 / 40 = 0.625
P(Weather = Good | −) = 55 / 60 = 0.9166
P(Construction = Yes | +) = 15 / 40 = 0.375
P(Construction = Yes | −) = 20 / 60 = 0.333

(b) Use the class conditional probabilities given in the previous question to predict the class label of a test example with the following feature set: (Accident = no, weather = bad, construction = yes) by applying the naive Bayes classifier.

P(+) = 40/100 = 0.4 , P(−) = 60/100 = 0.6

+=P(+)xP(Accident=No | +)xP(Weather=Bad | +)xP(Construction=Yes | +) = 0.4 x (15/40) x (15/40) x (0.375) = **0.02109375**

−=P(−)xP(Accident=No | −)xP(Weather=Bad | −)xP(Construction=Yes | −) = 0.6 x (50/60) x (5/60) x0.333 = **0.013875**

Class = POSITIVE (Congestion = Yes)

3. (6 points) Consider the following dataset.

Record	A	B	C	Class
1	0	0	0	+
2	0	0	1	-
3	0	1	1	-
4	0	1	1	-
5	0	0	1	+
6	1	0	1	+
7	1	0	1	-
8	1	0	1	-
9	1	1	1	+
10	1	0	1	+

(a) Estimate the conditional probabilities for the following:

$$P(A \mid +), \quad P(B \mid +), \quad P(C \mid +), \quad P(A \mid -), \quad P(B \mid -), \quad P(C \mid -).$$

P(A | +) = 3/5 = 0.6
P(B |+) = 1 / 5 = 0.2
P(C |+) = 4/5 = 0.8
P(A|−) = 2 / 5 = 0.4
P(B |−) = 2 / 5 = 0.4
P(C |−) = 5/5 = 1

(b) Using the conditional probability estimates obtained in part (a), predict the class label for the following test sample using the **Naive Bayes classifier**:

$(A = 0,\ B = 1,\ C = 0)$.

+ = P(+) x P(A=0 | +) x P(B=1 | +) x P(C=0 | +) =
(5/10) x (2/5) x (0.2) x (1/5) = **0.008**

- = P(-) x P(A=0 | -) x P(B=1 | -) x P(C=0 | -) =
(5/10) x (3/5) x (0.4) x 0 = **0**

Class = +

(c) Estimate the conditional probabilities using the **m-estimate** approach with the following parameters:

$p = \frac{1}{2}, \qquad m = 4.$

P(A | +) = (3 + 4 × 1/2) / (5 + 4) = 5/9
P(B | +) = (1 + 4 × 1/2) / (5 + 4) = 3/9
P(C | +) = (4 + 4 × 1/2) / (5 + 4) = 6/9
P(A=0 | +) = (2 + 4 × 1/2) / (5 + 4) = 4/9
P(B=0 | +) = (4 + 4 × 1/2) / (5 + 4) = 2/3
P(C=0 | +) = (1 + 4 × 1/2) / (5 + 4) = 1/3

P(A | -) = (2 + 4 × 1/2) / (5 + 4) = 4/9
P(B | -) = (2 + 4 × 1/2) / (5 + 4) = 4/9
P(C | -) = (5 + 4 × 1/2) / (5 + 4) = 7/9
P(A=0 | -) = (3 + 4 × 1/2) / (5 + 4) = 5/9
P(B=0 | -) = (3 + 4 × 1/2) / (5 + 4) = 5/9
P(C=0 | -) = (0 + 4 × 1/2) / (5 + 4) = 2/9

(d) Repeat part (b) using the conditional probabilities obtained in part (c).

+ = P(+) x P(A=0 | +) x P(B=1 | +) x P(C=0 | +) =
(5/10) x (4/9) x (3/9) x (1/3) = **0.024691**

- = P(-) x P(A=0 | -) x P(B=1 | -) x P(C=0 | -) =
(5/10) x (5/9) x (4/9) x 2/9 = **0.027434**

Class = -

(e) Compare the two probability estimation methods (maximum likelihood estimation and the m-estimate). Which method is preferable and why?

Compare :

1. Maximum likelihood P = nc / n :
   - Uses only real data
   - Can produce zero probabilities
   - Overfits small datasets

2. M-estimate P = (nc + m×p) / (n + m) :
   - Uses real data + prior knowledge
   - Never produces zero probabilities
   - Smooths extreme estimates
   - More robust with small datasets

The m-estimate is preferable because maximum likelihood gave P(C=0 | -) = 0 which caused - = 0 in part b and made the negative class impossible to predict. The M-estimate provides more realistic and robust probability estimates by avoiding zeros and smoothing toward the prior p = 0.5.

4. (5 points) Compare eager classification (decision trees, Bayesian classifiers, neural networks) with lazy classification (k-nearest neighbor).

Eager Learning (Decision Trees, Bayesian, Neural Networks):

- Builds a model during training phase
- Long training time, fast prediction time
- Small memory (stores only model)
- Must retrain when new data arrives
- Global decision boundary

Lazy Learning (k-Nearest Neighbors):

- Stores training data, no model built
- Short training time, slow prediction time
- Large memory (stores all training data)
- Easily adds new data (no retraining)
- Local decision boundary (can be arbitrary shapes)